

SIR SYED UNIVERSITY OF ENGINEERING & TECHNOLOGY
COMPUTER SCIENCE & INFORMATION TECHNOLOGY DEPARTMENT

Spring 2026
Software Quality Assurance & Testing (CS-468)
Assignment # 3

Semester: VIII

Announced Date: June 04, 2026

Total Marks: 4

Instructor Name: Farhana Ashraf

Batch: 2022F

Due Date: Sec A Jun 18, 26/Sec B Jun 16, 26/Sec C Jun 17, 26/Sec D Jun 15, 26

Marks Obtained:

- Instructions:**
- Attempt all questions.
 - Submit your handwritten assignment in form of hardcopy in person between 10:30am to 01:30pm
 - A viva would be conducted at the time of submission.

CLO #	Course Learning Outcomes (CLOs)	PLO Mapping	Bloom’s Taxonomy
CLO 2	Write and Use test cases to evaluate functional and non-functional specification requirements for various test types /levels / techniques and types of information systems and technologies.	PLO_4 (Design/ Development of Solutions)	C3 (Applying)

Key to solve the given code to answer upcoming questions:

1. Your Roll Number comprised of three things: Batch. Discipline and Roll No. In the code given below, By ‘Roll_No’ means the last digits of your Roll Number; it does not include batch and discipline. For example, if your Roll No, is 2022F-BCS-001 then you will consider 1 in ‘Roll_No’. Similarly, if your Roll No, is 2022F-BCS-011 then you will consider 11 in ‘Roll_No’ and if your Roll No, is 2022F-BCS-111 then you will consider 111 in ‘Roll_No’ to answer the upcoming questions.
2. If:
- a. Last three digits of your Roll number is an even number and it lies between 1 to 99 then replace line number 10 to 13 with ‘while loop’ and replace line number 20 to 23 with do-while loop.

b. Last three digits of your Roll number is an odd number and it lies between 1 to 99 then replace line number 10 to 13 with ‘do-while loop’ and replace line number 20 to 23 with while loop.

c. Last three digits of your Roll number is an even number and it lies between 100 to 199 then replace line number 10 to 13 with ‘for’ loop and replace line number 20 to 23 with repeat-until loop.

d. Last three digits of your Roll number is an odd number and it lies between 100 to 199 then replace line number 10 to 13 with ‘repeat-until’ loop’ and replace line number 20 to 23 with ‘for’ loop.

e. Last three digits of your Roll number is an ~~odd~~ even number and it lies between 200 to 300 then replace line number 10 to 13 with ‘repeat-until’ loop’ and replace line number 20 to 23 with ‘while’ loop.

f. Last three digits of your Roll number is an odd number and it lies between 200 to 300 then replace line number 10 to 13 with ‘while loop’ and replace line number 20 to 23 with ‘repeat-until’ loop.

WHILE LOOP	DO-WHILE LOOP
I=0 WHILE I<10 DO PRINT A END WHILE	I=0 WHILE I<10 DO PRINT A END WHILE
FOR LOOP	REPEAT UNTIL LOOP
FOR I=0 TO 9 PRINT A END FOR	REPEAT PRINT A UNTIL I<10

Given code to answer the following questions. Please read above key first and proceed accordingly to solve the questions.

```
1. READ Roll_No
2. A=Roll_No*-1
3. B=Roll_No*-10
4. C=Roll_No*50
5. X=C+A-B
6. Y=C+B-A
7. Z=A+B+C
8. IF X>Y THEN
9.     IF X>Z THEN
10.         I=0
11.         WHILE I<10 DO
12.             PRINT A
13.         END WHILE
14.         PRINT B
15.     ELSE
16.         PRINT C
17.     END IF
18. ELSE
19.     IF Y>Z THEN
20.         I=0
21.         WHILE I<10 DO
22.             PRINT A
23.         END WHILE
24.         PRINT B
25.     ELSE
26.         PRINT C
27.     END IF
28. END IF
```

Q1.

[4 marks]

For the program given above answer following.

- a. **Construct** a control flow graph.
- b. **Construct** a hybrid flow graph.
- c. **Write** test data for maximum statement coverage.
- d. **Write** test data for maximum decision coverage.
- e. **Discover** how many test cases would be needed for 100 percent statement coverage?
- f. **Discover** how many test cases are needed to achieve 100 percent decision coverage?
- g. **Compute** what level of decision coverage would be achieved by the single input Roll_No is equal to last three digit of your roll number?
- h. **Compute** whether 100% statement coverage is achieved if Roll_No is equal to last three digit of your roll number? If not, **identify** which lines of pseudo code will not be exercised?

